

3 (a) evidence of use of area below the line distance = 39 m (<i>allow ± 0.5 m</i>) (<i>if $> \pm 0.5$ m but ≤ 1.0 m, then allow 1 mark</i>)	B1 A2	[3]
(b) (i) 1 $E_K = \frac{1}{2}mv^2$ $\Delta E_K = \frac{1}{2} \times 92 \times (6^2 - 3^2)$ = 1240 J	C1 A1	[2]
2 $E_P = mgh$ $\Delta E_P = 92 \times 9.8 \times 1.3$ = 1170 J	C1 A1	[2]
(ii) $E = Pt$ $E = 75 \times 8$ = 600 J	C1 A1	[2]
(c) (i) energy = $(1240 + 600) - 1170$ = 670 J	M1 A0	[1]
(ii) force = $670/39 = 17$ N	A1	[1]
(d) frictional forces include air resistance air resistance decreases with decrease of speed	B1 B1	[2]

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4 (a) (i) solid has fixed volume and fixed shape/incompressible

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[1]